Behavioral Assertion and Regression Toolkit

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The Architecture for the Digital World®
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Target Use Cases

- Kernel Developers
- Performance Engineers
- Release/QA Engineers

The question?

How do we "test behaviours"?

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Trace Events to DataFrames

Unique Word Key-Value Pairs

```
kworker/6:1-459   [006]  2806.211584: thermal_temperature:      thermal_zone=exynos-therm id=0 temp_prev=72349 temp=72221
kworker/6:1-459   [006]  2806.211712: thermal_power_cpu_limit: cpus=000000f0 freq=2200000 cdev_state=0 power=4567
kworker/6:1-459   [006]  2806.211718: cdev_update:             type=thermal-cpufreq-1 target=0
```

ThermalTemperature

CpuPower

CdevUpdate

pandas.DataFrame

	__comm	__cpu	__pid	id	temp	temp_prev	thermal_zone
Time							
0.000000	kworker/6:1	6	459	0	50054	50054	exynos-therm
0.999983	kworker/6:1	6	459	0	50574	50054	exynos-therm
2.000008	kworker/6:1	6	459	0	53323	50574	exynos-therm
2.999933	kworker/6:1	6	459	0	52239	53323	exynos-therm
3.999964	kworker/6:1	6	459	0	54564	52239	exynos-therm

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Grammar

Event : Column



Pivoting
Select output for tz_id = 0

```
getStatement("mean(THERMAL:temp - THERMAL:temp_prev)", select=0)

getStatement("(std(THERMAL:temp) * 100.0) / mean(THERMAL:temp)", select=0)

assertStatement("mean(THERMAL:temp") < CONTROL_TEMP",  select=0)
```


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Example Assertion: DynamicPower

$$\sum_{i=0}^{cpus} Load_i(t) = 0 \implies dynamic\ power(t) = 0$$

```
getStatement(    "(\
                   (IN:load0 + IN:load1 + IN:load2 + IN:load3) == 0\
                  )\
                   \
                  & (IN:dynamic_power == 0)",

                  select=CLUSTER_0
             )
```


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Thermal Behaviors

DEMO

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Plotting

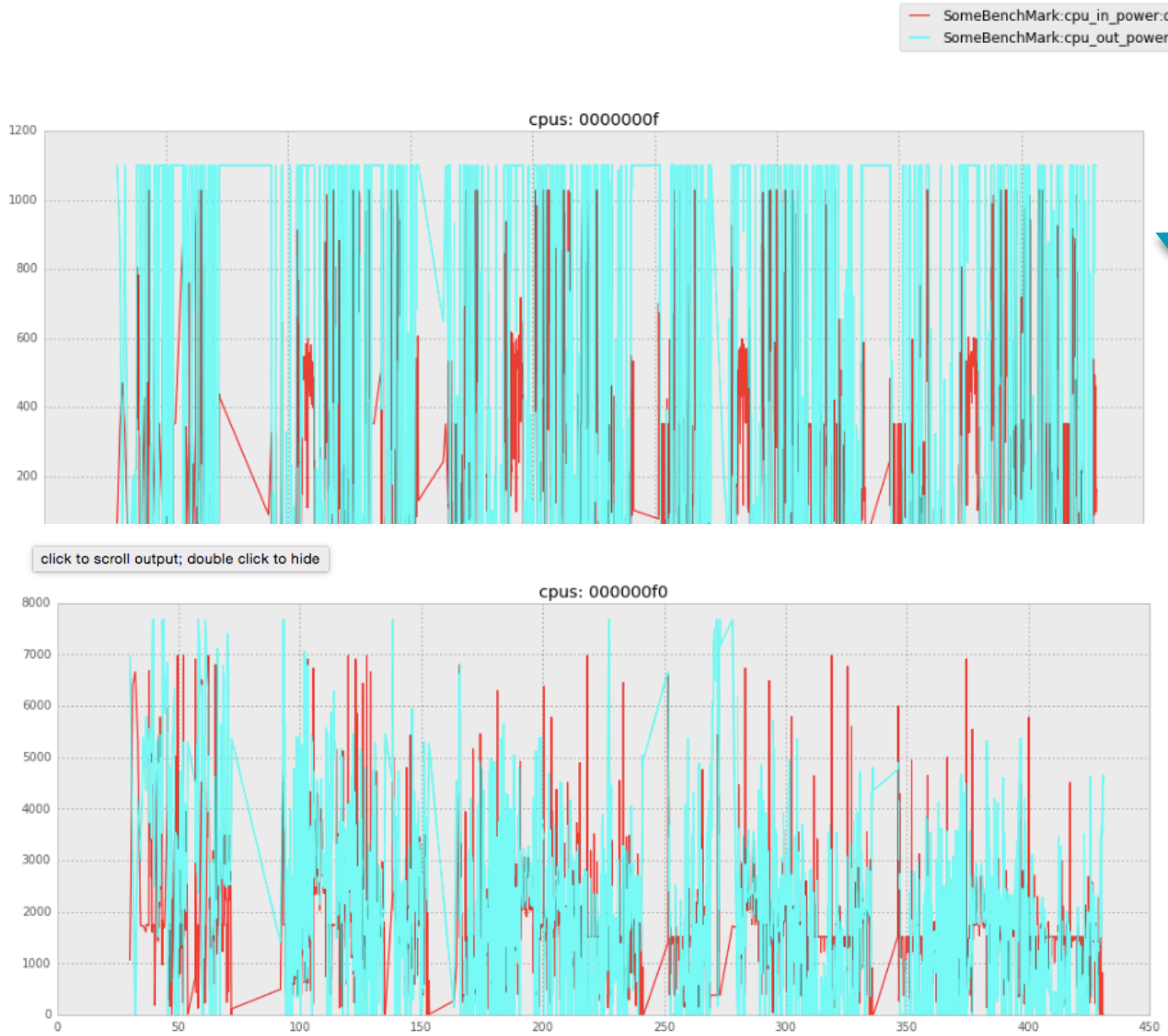
Multiple Runs, Columns and Events Supported




```
ILinePlot(run,
          [
              CpuInPower,
              CpuOutPower
          ],
          column=[ "dynamic_power",
                   "power"
                 ],
          pivot="cpus"
          )
```


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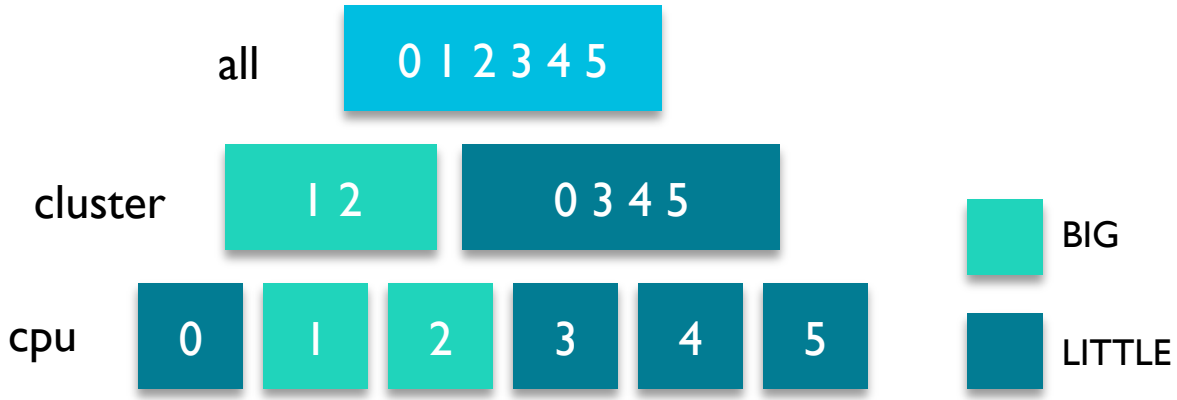
Static and Interactive Plots



One Plot per Pivot

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Scheduler Assertions: Vector



CPU TOPOLOGY

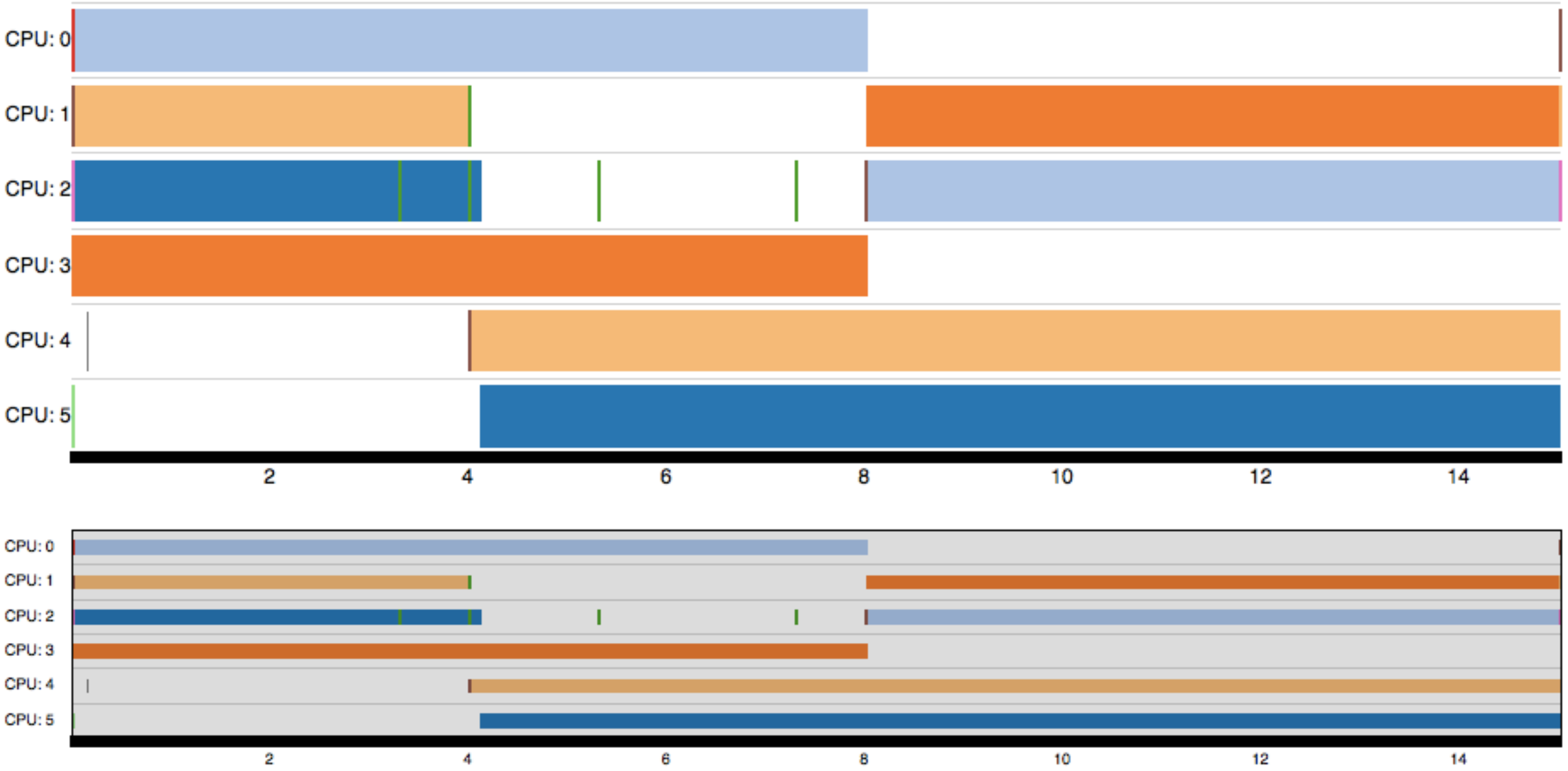
```
BIG = [1, 2]
LITTLE = [0, 3, 4, 5]
clusters = [BIG, LITTLE]


topology = Topology(clusters=clusters)

s = SchedMultiAssert(trace_dir, topology, execnames="sysbench")
```


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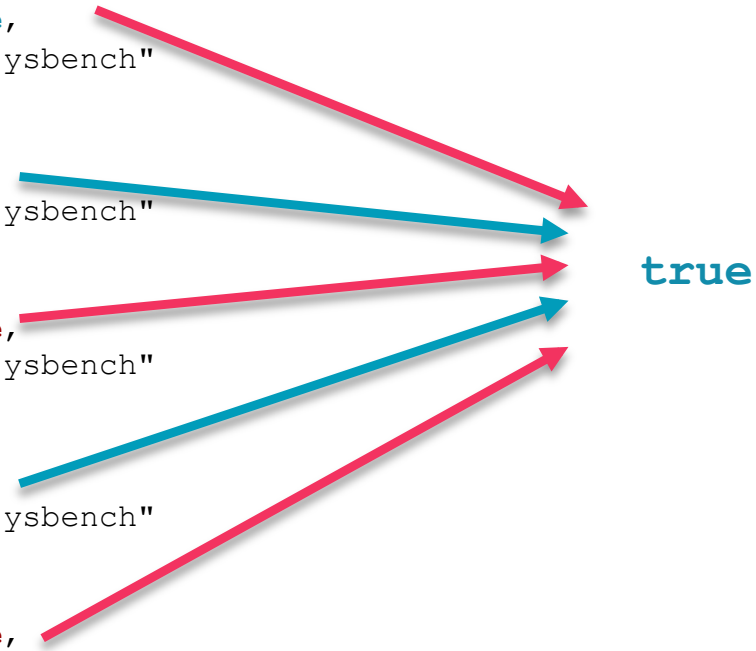
Trace under Analysis



assertSwitch

```
assertSwitch("cluster", BIG, LITTLE, window=(3.8, 4.2),
             rank=2)
```

```
{
   "1825": {
      "switch": false,
      "task_name": "sysbench"
   },
   "1826": {
      "switch": true,
      "task_name": "sysbench"
   },
   "1827": {
      "switch": false,
      "task_name": "sysbench"
   },
   "1828": {
      "switch": true,
      "task_name": "sysbench"
   },
   "1823": {
      "switch": false,
      "task_name": "sysbench"
   }
}
```

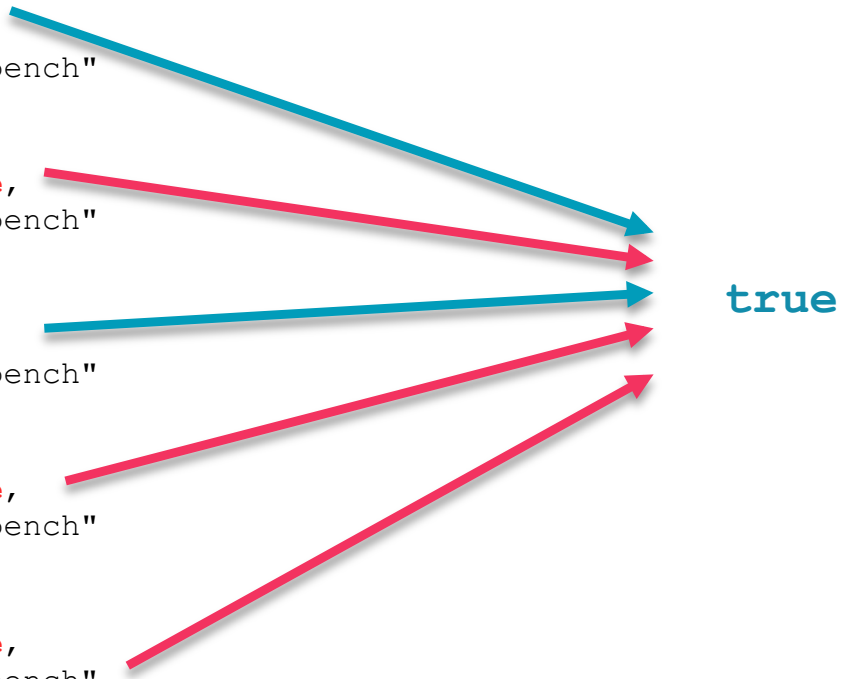


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assertResidency

```
assertResidency("cluster", BIG, 45, between_threshold, percent=True
                rank=2)
```

```
{
    "1825": {
        "residency": true,
        "task_name": "sysbench"
    },
    "1826": {
        "residency": false,
        "task_name": "sysbench"
    },
    "1827": {
        "residency": true,
        "task_name": "sysbench"
    },
    "1828": {
        "residency": false,
        "task_name": "sysbench"
    },
    "1823": {
        "residency": false,
        "task_name": "sysbench"
    }
}
```




```
def between_threshold(a, b):

    threshold = 5
    if abs(a - b) > threshold:
        return False
    else:
        return True
```


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getResidency

```
getResidency("cluster", BIG, percent=True)
```


```
{
   "1825": {
      "residency": 46.548574941586658,
      "task_name": "sysbench"
   },
   "1826": {
      "residency": 26.71621003370441,
      "task_name": "sysbench"
   },
   "1827": {
      "residency": 46.535288658721988,
      "task_name": "sysbench"
   },
   "1828": {
      "residency": 27.459386427327228,
      "task_name": "sysbench"
   },
   "1823": {
      "residency": 96.405447644712282,
      "task_name": "sysbench"
   }
}
```


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Scheduler Behaviors

DEMO

SCHED DEADINE

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Scheduler Metrics and Users

Period

Duty Cylcle

Run Time

First CPU

Current Users

Sched Deadline

Energy Aware + CFS

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